Title & Executive Summary

OVERVIEW — Focused Scope: AI Data Center Infrastructure

99.9999%	$2.4M / hr	5 Top Tier
Target Uptime	Annual Downtime Cost Risk	Key Vendors Analyzed

Executive Summary & Overview:

Executive Study context updated per new problem statement: 'Evaluate Power Quality Monitoring for Hyperscale AI Data Center Expansion'. Power Quality Monitoring (PQM) represents a critical operational commodity for advanced manufacturing, semiconductor fabrication, and enterprise data centers. Voltage sags, harmonic distortions, and transient spikes directly threaten capital equipment and production yield. This executive study evaluates market dynamics, key suppliers, technology architectures, and selection criteria to optimize procurement and technical program management. (Revised per Executive Feedback: Instead of Tile & Executive Summary - use Power Quality Monitoring (PQM) Executive Study)

Key Takeaways:

- Power Quality Monitoring (PQM) is essential for uptime in high-density facilities, Semiconductor Fabs, and Data Centers. [Updated: Instead of Tile & Executive Summary - us...]
- Rapid growth driven by AI workload density, microgrids, and strict utility compliance standards.
- Evaluation framework targets top global players: Eaton, Schneider Electric, SEL, Janitza, and Powerside.

PQM 1 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **EPRI Power Quality Basics**: https://www.epri.com/research/products/000000003002016480 | **IEEE 1159 Monitoring Practice**: https://standards.ieee.org/ieee/1159/5443/

Commodity Definition & Industry Relevance

COMMODITY SPECS — Physics of Power Anomalies & Impact on Critical Infrastructure

IEC 61000-4-30 Ed 3	50% Sag Threshold	Up to 63rd Harmonic
Class A Accuracy	SEMI F47 Immunity	Harmonic Spectrum

Executive Summary & Overview:

The PQM commodity encompasses hardware meters, transient recorders, CT/PT transducers, and edge analytics software. In semiconductor wafer fabs, even a 12-cycle voltage sag can trip automated track systems and destroy $10M+ in work-in-progress. PQM hardware serves as the primary diagnostic sensor array to detect, capture, and isolate electrical anomalies before catastrophic failure occurs. (Revised per Executive Feedback: Add a glossary slide which explains the acronyms like PQM, CP/PT etc)

Key Takeaways:

- Power disturbances include voltage sags/swells, total harmonic distortion (THD), transients, and flicker. [Updated: Add a glossary slide which explains the ...]
- SEMI F47 compliance mandates tool immunity to voltage sags down to 50% for up to 1000ms.
- Advanced Class A monitoring (IEC 61000-4-30) guarantees legally defensible utility data accuracy.

PQM 2 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **IEC 61000-4-30 Standard**: https://www.iec.ch/dyn/www/f?p=103:22:0::::FSP_ORG_ID,FSP_LANG_ID:1265,25 | **SEMI F47 Voltage Sag Standard**: https://www.semi.org/en/products-services/standards

Market Dynamics & Macro Drivers

MARKET TRENDS — TAM Expansion, AI Workload Power Surge & Grid Decarbonization

$4.2 Billion	6.8% (2024-2030)	+160% by 2030
Global Market TAM	Market CAGR	AI Grid Demand Growth

Executive Summary & Overview:

Macroelectric trends are accelerating PQM adoption. The explosion of AI data center compute requires unprecedented power density (100kW+ per rack), increasing susceptibility to thermal and electrical transients. Concurrently, industrial electrification and renewable energy integration add non-linear loads, driving mandatory installation of continuous power quality logging systems. (Revised per Executive Feedback: Add reference links to takeaways like where did you see the solar/wind microgrids introduce bi-directional harmonic distortion)

Key Takeaways:

- Global Power Quality Hardware & Software Market growing at 6.8% CAGR toward $4.2B by 2030. [Updated: Add reference links to takeaways like wh...]
- Hyperscale AI data centers demand high-density transient monitoring for megawatt-scale rack deployments.
- Distributed Energy Resources (DERs) and solar/wind microgrids introduce bi-directional harmonic distortion.

PQM 3 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **Grand View Research Market Analysis**: https://www.grandviewresearch.com/industry-analysis/power-quality-equipment-market | **IEA Electricity 2024 Report**: https://www.iea.org/reports/electricity-2024

Competitive Landscape & Market Share

COMPETITIVE INTELLIGENCE — Market Distribution Across Key Industry Players

32% Market Share	23% Market Share	45% Combined Market
Schneider Electric	Eaton Corporation	SEL / Janitza / Powerside

Executive Summary & Overview:

The PQM supplier ecosystem is divided between enterprise power infrastructure conglomerates and specialized power quality measurement pioneers. Schneider Electric and Eaton offer broad EcoStruxure/Brightlayer software suites. SEL brings utility-grade protection reliability, while Janitza and Powerside offer specialized high-sampling edge devices with rapid deployment capabilities.

Key Takeaways:

- Tier 1 Giants (Schneider Electric & Eaton) control ~55% of global enterprise power monitoring systems.
- SEL dominates utility substation & industrial protective relay integration (~18% market share).
- Niche innovators (Janitza, Powerside) lead high-precision edge meters and modular micro-PQ technology (~15%).

PQM 4 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **Schneider EcoStruxure Power**: https://www.se.com/ww/en/work/solutions/system-availability/ | **Eaton Brightlayer Power Solutions**: https://www.eaton.com/us/en-us/digital/brightlayer.html

Supplier Differentiation & USPs

VENDOR MATRIX — Deep Dive Comparison: Strengths, Weaknesses & USPs

> 500,000 hrs	1024 samples/cycle	Up to 14 channels
SEL Hardware MTBF	ION9000 Sampling Rate	PQube3 Channels

Executive Summary & Overview:

Selecting the right supplier depends on system architecture. Enterprise facilities requiring centralized software lean toward Schneider ION series or Eaton. Mission-critical utility environments favor SEL's bulletproof hardware. Industrial machine builders and modular data center ops favor Janitza and Powerside for open interfaces and compact footprints. (Revised per Executive Feedback: Add a reference to how each supplier has achieved its differentiation)

Key Takeaways:

- Schneider (ION9000): Industry benchmark for high-speed waveform capture and EPMS integration. [Updated: Add a reference to how each supplier has...]
- Eaton (Brightlayer): Deep hardware catalog with native UPS & PDU software integration.
- SEL (SEL-735): 10-year warranty, ultra-rugged substation grade, unmatched MTBF (>500k hours).
- Janitza (UMG Series): Open Modbus/REST APIs, compact DIN-rail form factors, native web servers.
- Powerside (PQube3): Ultra-compact form factor, auto-configured report generation, plug-and-play.

PQM 5 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **SEL Power System Products**: https://selinc.com/products/ | **Powerside PQube3 Technical Specs**: https://powerside.com/products/pqube-3/

Product & Hardware Architecture

HARDWARE ARCHITECTURE — Sensor Classifications, Sampling Frequency & Edge Data Capture

±1 µs	20 MHz Sampling	CT, PT, Rogowski
GPS Sync Accuracy	Transient Resolution	Transducer Types

Executive Summary & Overview:

PQM hardware architecture combines analog-to-digital converters (ADCs), digital signal processors (DSPs), non-volatile memory, and Ethernet/fieldbus communications. High-end meters feature dual-Ethernet ports for segregated operational technology (OT) networks and support PTP (Precision Time Protocol) for microsecond event correlation across global multi-site installations. (Revised per Executive Feedback: Make the acronyms linked to glossary slide for example on this slide if executive wants to see what is GPS/PTP or know more about Split-core Rogowski coils)

Key Takeaways:

- Class A meters (IEC 61000-4-30 Ed 3) require synchronized clock accuracy (GPS/PTP) to ±1 microsecond. [Updated: Make the acronyms linked to glossary sli...]
- High-speed transient capture ranges from 20 kHz to 5 MHz for microsecond impulse detection.
- Split-core Rogowski coils enable non-intrusive retrofit installations without facility shutdown.

Software Platforms & Power Analytics

SOFTWARE ARCHITECTURE — EPMS Architectures, Cloud Dashboards & AI Anomaly Detection

< 100 ms	Modbus, IEC 61850, MQTT	94.2% Anomaly Flag
Analytics Latency	Standard Protocols	Predictive Accuracy

Executive Summary & Overview:

Hardware value is unlocked through software platforms like Schneider Power Monitoring Expert (PME) and Eaton Foreseer. Modern EPMS platforms combine real-time single-line diagrams, event disturbance direction detection (upstream vs downstream fault origin), and compliance report automation (EN 50160, IEEE 519). (Revised per Executive Feedback: Add a visual for the CBEMA/ITIC curve analysis and details for the impact of not being able to detect transformer insulation breakdown)

Key Takeaways:

- Electrical Power Management Systems (EPMS) aggregate millisecond-level telemetry across thousands of nodes. [Updated: Add a visual for the CBEMA/ITIC curve an...]
- Automated CBEMA / ITIC curve analysis plots voltage sags to immediately establish equipment tolerance compliance.
- AI/ML predictive analytics detect transformer insulation breakdown and impending harmonic resonance.

Service Offerings & Lifecycle Management

LIFECYCLE & SERVICES — Commissioning, NIST Traceable Calibration & Cybersecurity Protocols

IEC 62443-4-2	Annual / Biennial	< 2 Hours
Cyber Standard	Calibration Interval	SLA Response Time

Executive Summary & Overview:

Lifecycle stewardship goes beyond initial hardware installation. Enterprise procurement contracts must mandate vendor-backed commissioning, relay coordination studies, annual calibration, and firm software security maintenance agreements to protect critical power infrastructure against physical and cyber threats.

Key Takeaways:

- Annual NIST-traceable calibration required to preserve legal defensibility in utility tariff disputes.
- IEC 62443-4-2 cybersecurity compliance mandatory for OT network-connected meters.
- Vendor SLAs cover firmware hot-patching, 24/7 technical support, and hardware replacement guarantees.

PQM 8 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **ISA/IEC 62443 Cybersecurity Standard**: https://www.isa.org/standards-and-publications/isa-standards/isa-standards-committees/isa99 | **NIST Cybersecurity Framework**: https://www.nist.gov/cyberframework

Vendor Evaluation & Selection Criteria

PROCUREMENT STRATEGY — Weighted Decision Matrix for Procurement & Engineering Alignment

30%	25%	20%
Technical Weight	TCO & Cost Weight	Interoperability Weight

Executive Summary & Overview:

To eliminate bias in commodity vendor selection, technical program leaders implement a multi-attribute utility matrix. Points are awarded based on independent laboratory certifications, protocol interoperability, firmware vulnerability management, total cost of ownership over a 10-year lifespan, and local field support capabilities.

Key Takeaways:

- Evaluation model balances Technical Performance (30%), TCO (25%), Interoperability (20%), Cyber/Security (15%), and Vendor Resilience (10%).
- Open protocol support (Modbus TCP, IEC 61850, REST APIs) prevents proprietary software lock-in.
- Hardware modularity enables rapid field replacement without re-wiring CT/PT lines.

PQM 9 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **IEC 61850 Communication Standard**: https://www.iec.ch/grid/smartgrid/smartgrid_61850.htm | **Modbus Organization Specifications**: https://modbus.org/specs.php

Total Cost of Ownership (TCO) & Sourcing Risk

TCO & RISK — CapEx vs OpEx Breakdown & Supply Chain Resilience Strategy

35%	65%	16-24 Weeks
Hardware CapEx Share	Software & Services Share	Avg Hardware Lead Time

Executive Summary & Overview:

Procurement strategies must account for total cost of ownership and hardware lead-time risk. While Tier 1 meter hardware carries a premium, their long lifecycle support lowers long-term OpEx. Second-sourcing with open-standard meters (e.g. Janitza or Powerside) provides critical supply chain flexibility for accelerated NPI facility builds. (Revised per Executive Feedback: Make the costs and lead times drillable to the next level below)

Key Takeaways:

- Hardware CapEx represents only 35% of 10-year TCO; installation, software licensing, and maintenance comprise 65%. [Updated: Make the costs and lead times drillable ...]
- Lead-time volatility (16-32 weeks for specialized meters) necessitates strategic inventory buffering.
- Multi-sourcing critical meter points avoids single-vendor reliance during supply chain disruptions.

PQM 10 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **Gartner Supply Chain Procurement**: https://www.gartner.com/en/supply-chain | **McKinsey Supply Chain Resilience**: https://www.mckinsey.com/capabilities/operations/our-insights

Implementation & Operational Roadmap

EXECUTION ROADMAP — Phased NPI Deployment, Clear-To-Build (CTB) & FRACAS Governance

Weeks 1-4	Weeks 5-12	Weeks 13-24
Phase 1 Audit	Phase 2 Pilot Ramp	Phase 3 Full Rollout

Executive Summary & Overview:

Flawless deployment requires a disciplined NPI program management structure. Clear-To-Build (CTB) tracking ensures all CTs, PTs, power meters, and network gateways are staged prior to shutdown windows. Incorporating a Failure Reporting, Analysis, and Corrective Action System (FRACAS) ensures meter maintenance issues are systematically analyzed and corrected. (Revised per Executive Feedback: FRACAS is typically engaged AFTER a failure event happens, if such data is not available, show a FMECA - with potential failure modes and effects and what can be done to mitigate those risks)

Key Takeaways:

- Phase 1: Facility audit, single-line diagram mapping, and OT network architecture definition. [Updated: FRACAS is typically engaged AFTER a fail...]
- Phase 2: Pilot deployment & Clear-To-Build (CTB) verification across high-risk electrical sub-stations.
- Phase 3: Full-scale rollout, FRACAS governance setup, and automated EPMS integration.

PQM 11 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **ASQ FRACAS Quality Standard**: https://asq.org/quality-resources/fracas | **PMI Standard for Program Management**: https://www.pmi.org/pmbok-guide-standards

Strategic Recommendations & Sourcing Plan

RECOMMENDATIONS — Executive Action Plan & Dual-Vendor Sourcing Framework

14-18% TCO Target Cost Savings	**-30%** Deployment Lead Reduction	**Dual-Tier Supplier** Vendor Diversification

Executive Summary & Overview:

The optimal PQM sourcing strategy balances enterprise software sophistication with hardware resilience and open flexibility. By establishing a dual-vendor architecture, technical leaders can negotiate competitive contract pricing, minimize lead-time bottlenecks, and ensure 99.9999% power reliability across critical manufacturing and computing facilities. (Revised per Executive Feedback: Add what exactly will be done in engaging in a dual tier supplier strategy, cost and potential issues) (Revised per Executive Feedback: Add metrics for what must go right and thresholds for what can go wrong that must be monitored)

Key Takeaways:

- Adopt Dual-Vendor Sourcing: Schneider/SEL for main switchgear; Janitza/Powerside for machine-level sub-metering. [Updated: Add what exactly will be done in engagin...] [Updated: Add metrics for what must go right and t...]
- Mandate open protocol compliance (Modbus TCP / IEC 61850) in all RFPs to prevent lock-in.
- Establish centralized EPMS software to aggregate multi-site telemetry for proactive energy optimization.

PQM 12 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **IEEE Power & Energy Society**: https://www.ieee-pes.org/ | **CIGRE Power Quality Working Group**: https://www.cigre.org/

Next Steps & Implementation Execution Plan

EXECUTION PLAN — Key Deliverables, Timeline Milestones & Resource Budgeting

16 Weeks	$450,000	$1.8M / Year
Est. Execution Timeline	Total Implementation Cost	Expected Annual Savings

Executive Summary & Overview:

Execution plan aligned with problem requirements: 'Evaluate Power Quality Monitoring for Hyperscale AI Data Center Expansion...'. To execute the PQM commodity strategy, a 16-week phased implementation plan is established with a total budgeted cost of $450,000 (covering hardware procurement, commissioning, and EPMS integration). Key deliverables focus on vendor contract finalization, pilot installation across critical sub-stations, and establishing automated FRACAS governance to capture an estimated $1.8M in annual downtime risk avoidance.

Key Takeaways:

- Deliverable 1: Final RFP & Dual-Vendor Contracting (Schneider/SEL + Janitza/Powerside) [Weeks 1-4].
- Deliverable 2: Phase 1 Pilot Installation & CTB Staging across 2 primary substations [Weeks 5-10].
- Deliverable 3: Full EPMS Integration, FRACAS Governance & Executive Sign-off [Weeks 11-16].

PQM 13 | **PQM_Executive_Deck.pdf** | Author: **Lalit Sabnani** | Date: 2026-10-10 | Verified Sources: **PMI Project Budgeting & Governance Standard**: https://www.pmi.org/pmbok-guide-standards | **Gartner IT & Facilities Execution Framework**: https://www.gartner.com/en/supply-chain